Spatial Characterization of Tumor-Associated Macrophage Polarization and Anti-PD-1 Drug Uptake in a CT26 Colorectal Cancer Xenograft Model

Submitted by

Nasha Palsetia

Northeastern University

Boston, MA

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A. Specific Aims:

Precision oncology has transformed cancer treatment by targeting specific molecular and genetic alterations driving a patient's tumor, demonstrating improved clinical outcomes compared to traditional approaches. Despite this progress, the efficacy of targeted immunotherapies, such as anti-PD-1, varies significantly across patients with colorectal cancer (CRC), suggesting that factors within the tumor microenvironment influence therapeutic response. Tumor-associated macrophages (TAMs), one of the most abundant immune cells in the CRC microenvironment, exist along a polarization spectrum between pro-inflammatory M1 and immunosuppressive M2 states, with M2 macrophages causing immune evasion through PD-L1 upregulation. However, the spatial distribution of macrophage polarization states and their direct relationship to anti-PD-1 drug uptake has not been characterized in CRC, representing a critical gap in understanding localized resistance mechanisms.

The long-term goal of this research is to establish a spatial biology framework to characterize how tumor-associated macrophage polarization influences anti-PD-1 drug uptake and resistance in CRC. I hypothesize that regions of dominant M2 macrophage polarization within the CT26 CRC tumor microenvironment spatially correspond to reduced anti-PD-1 uptake, while regions of dominant M1 polarization correspond to greater drug distribution and therapeutic efficacy.

Specific Aim 1: To determine whether the spatial distribution of macrophage polarization states, characterized by CD68 and CD163 immunohistochemical (IHC) staining, correlates with regions of variable anti-PD-1 drug uptake in a CT26 colorectal cancer xenograft model. Radiolabeled anti-PD-1 will be administered to CT26 tumor-bearing BALB/c mice [1]. Tumor sections will be exposed to autoradiography to map drug distribution, then fixed in methanol and stained with CD68 and CD163 IHC to identify macrophage polarization states. The two images will be co-registered using ImageJ to determine whether macrophage polarization spatially corresponds to variable drug uptake [5].

Specific Aim 2: To determine whether PD-L1 expression, a functional marker of macrophage-mediated immunosuppression, spatially corresponds to regions of low anti-PD-1 drug uptake in a CT26 CRC xenograft model [10]. Building on Aim 1, IHC staining for PD-L1 will be performed on the same tumor sections, quantified using ImageJ, and co-registered with autoradiography drug uptake maps to determine whether areas of high PD-L1 expression correspond to regions of low drug uptake and resistance.

Altogether, this research aims to understand how the spatial composition of tumor-associated macrophages drives variability in anti-PD-1 drug uptake in CRC, providing a foundational framework to improve the design and efficacy of targeted immunotherapy for colorectal cancer patients.

B. Significance:

Colorectal Cancer & Targeted Therapy:

Colorectal cancer (CRC) is the third most common cancer type and the second leading cause of cancer-related deaths worldwide, affecting the colon (large intestine) or rectum. In 2022 alone, there were approximately 1.9 million new CRC cases and over 900,000 deaths globally. This type of cancer can develop from genetic and epigenetic modifications, as well as lifestyle factors, such as diet, obesity, tobacco use, and alcohol consumption, family and personal history, and age. Depending on the stage of disease and tumor characteristics, treatments include surgery, radiation therapy, chemotherapy, immunotherapy, and targeted therapy [2].

Targeted therapy has become an emerging approach to CRC and other cancer treatments, focusing on targeting specific molecular and genetic alterations driving a patient's tumor. By tailoring treatments to the molecular profile of tumors, targeted therapy has demonstrated improved clinical outcomes and fewer side effects compared to traditional approaches such as chemotherapy and radiation. Certain targeted therapies for CRC target growth factor receptors, vasculature proteins, genes, and growth-promoting proteins found in tumors [3]. However, the efficacy of targeted therapies varies significantly across patients, even within the same tumor type, suggesting that factors within the tumor microenvironment may influence treatment response [8]. Despite the promise of anti-PD-1 immunotherapy, response rates remain limited in CRC. The overall objective response rate of anti-PD-1 therapy in advanced CRC is approximately 23%, with response rates as low as 11% in microsatellite stable tumors, which represent the majority of CRC cases [9]. This limited and variable response underscores the need to better understand the tumor microenvironmental factors that influence drug uptake and resistance.

Macrophages in Tumor Microenvironments:

Macrophages are immune cells derived from bone marrow precursors and are present throughout the body, including in tissue and tumor microenvironments. Macrophages are classified into two types based on their phenotypic patterns and functional properties [11].
The M1-type macrophages are pro-inflammatory, phagocytic cells that provide effective defense against pathogens and support anti-tumor immunity. These macrophages secrete chemokines and activate T-cell function against cancers. In contrast, the M2-type macrophages are anti-inflammatory, pro-angiogenic cells with tissue repair and tumor immune evasion properties. These macrophages specifically release cytokines and growth factors to assist tumor growth [7].

While the two types of macrophages have polarizing functions, tumor-associated macrophages (TAMs) exist along a polarization spectrum between M1 and M2 states, with high plasticity that allows phenotypic shifts in response to microenvironmental signals [4].

Gap in Knowledge:

While TAMs are one of the most abundant immune cells in CRC microenvironments, their role in targeted therapy resistance remains poorly understood. Although M2 macrophages have been associated with immune evasion through the upregulation of the ligand PD-L1 and suppression of T-cell activity, the spatial distribution of M1 and M2 macrophage phenotypes within the tumor and their direct relationship to the uptake of targeted therapies has not been thoroughly characterized [10]. Additionally, given the high plasticity of TAMs and their sensitivity to changes in microenvironments, understanding how macrophage polarization varies spatially across distinct tumor regions is critical to identify mechanisms of treatment resistance. Addressing this is essential for improving the design and efficacy of targeted therapies in CRC. Spatial biology has emerged as an approach to address this limitation, enabling molecular features to be mapped within their tissue context and preserving spatial relationships between cells that conventional sequencing approaches lose upon dissociation [6]. Applying this framework to characterize macrophage distribution in relation to drug uptake offers a novel approach to understanding localized resistance in CRC.

Proposed Research:

The research aims to characterize the spatial distribution of tumor-associated macrophages and their relationship to targeted therapy uptake in a CT26 CRC xenograft model. Autoradiography will be used to visualize and quantify the distribution of the anti-PD-1 drug across tumor regions, which will then be co-registered with immunohistochemical (IHC) staining to map M1 and M2 macrophage phenotypes and PD-L1 expression spatially.

I hypothesize that regions of dominant M2 macrophage polarization spatially correspond to reduced anti-PD-1 uptake and impaired treatment response, reflected by tumor volume measurements, while regions of dominant M1 polarization correspond to greater drug distribution and therapeutic efficacy. By observing the spatial relationship between macrophage polarization and drug uptake, this research can help address how the tumor immune microenvironment contributes to targeted therapy resistance in CRC. The long-term goal of this study is to identify macrophage-related biomarkers that predict treatment response and contribute to more effective precision oncology strategies for patients with CRC.

C. Innovations:

The proposed research introduces a novel spatial framework for understanding macrophage-mediated resistance to immunotherapy in colorectal cancer. While individual components of this approach exist independently, their integration to directly correlate immune polarization landscapes with radiolabeled drug distribution within the same tumor section has not been established in CRC. This combination uniquely positions the research to answer a question that neither technique could address alone: not simply whether macrophages influence resistance, but precisely where within the tumor that resistance is spatially occurring and whether it can be predicted by macrophage polarization state.

Spatial Biology Framework Providing Spatial Immune Context:

Conventional approaches to studying TAMs rely on bulk sequencing or flow cytometry, which lose critical spatial information when cells are dissociated from tissues. Recent advancements in spatial biology have enabled researchers to map molecular features within tissues, preserving the spatial relationships between immune cells and enabling a more precise characterization of the spatial immune landscape [6]. This study leverages this emerging framework to address not just whether macrophages influence resistance, but where in the tumor that resistance is occurring. While emerging technologies such as spatial transcriptomics and mass spectrometry imaging offer additional resolution, the proposed co-registration approach provides a practical and clinically accessible framework for spatially characterizing immune resistance in CRC.

Co-registration of Drug Uptake and Immune Mapping:

While autoradiography and IHC staining are individually established imaging techniques, their co-registration on the same tumor section for simultaneous spatial mapping of drug distribution and macrophage phenotypes represents a novel methodology in CRC research. The approach directly correlates immune microenvironmental features with drug distribution at a spatially resolved level, which neither image alone could accomplish [5]. By overlaying anti-PD-1 uptake maps with macrophage polarization and PD-L1 expression on the same tissue section, this methodology can characterize whether localized resistance patterns in CRC are spatially driven by macrophage polarization states.

Syngeneic CRC Tumor Model for Authentic Immune Cell Study:

Most xenograft models require immunocompromised hosts, eliminating the necessary immune components that the research aims to study. The CT26 syngeneic model uniquely preserves a fully functional immune system in the tumor, enabling authentic macrophage behavior and immune-drug interactions to be observed within the intact microenvironment. This model enables the proposed study to generate findings that are not only meaningful but also more clinically translatable than conventional immunocompromised models, positioning this research to spatially characterize immune-mediated resistance in a way no immunocompromised model could support [1].

D. Approach:

Specific Aim 1: To determine whether the spatial distribution of macrophage polarization states, characterized by CD68 and CD163 IHC staining, correlates with regions of variable anti-PD-1 drug uptake in a CT26 colorectal cancer xenograft model.

Methods and Rationale:

Female BALB/c mice aged 6-8 weeks will be subcutaneously inoculated with CT26.WT-luc colorectal cancer cells. Mice will be randomly assigned to two groups, an anti-PD-1 treatment group and an isotype control group, with a minimum of 5 animals per group to ensure adequate statistical power. Treatment will begin when tumors reach an average volume of 100-150mm^3, which is approximately 7-10 days post-implantation. Radiolabeled anti-PD-1 antibodies will be administered at 200μg per mouse [1]. Tumor volume will be monitored biweekly using caliper measurements to establish an efficacy endpoint. Tumors will be collected 24-48 hours following the final drug administration, frozen, and sectioned at 12 μm for imaging.

To address potential signal interference between autoradiography and IHC staining, tumor sections will first be exposed to autoradiography to capture drug distribution, then fixed in methanol prior to IHC staining. This sequential protocol preserves both the autoradiographic signal and tissue antigenicity, enabling reliable co-registration of drug uptake and immune cell distribution on the same section [7]. Autoradiography imaging will be performed to visualize and quantify the spatial distribution of anti-PD-1 drug uptake, with regions of highest and lowest uptake mapped using ImageJ. IHC staining for CD68 and CD163 will then be performed on the same section to identify M1 and M2 macrophages respectively. The macrophage polarization states will be quantified and co-registered with autoradiography findings to determine whether macrophage polarization corresponds to variable drug uptake. Co-registration allows direct spatial comparison of drug distribution and immune cell composition, reducing variability and improving data quality [5].

Anticipated results:

If the hypothesis is supported: Regions of dominant M1 polarization would correspond to areas of high anti-PD-1 drug uptake, while regions of dominant M2 polarization would correspond to areas of low drug uptake in the CT26 tumor model. This would support the hypothesis that the polarization of TAMs spatially predicts variability in drug uptake and suggests that regions enriched in M2 represent sites of resistance within the CT26 tumor.

If the hypothesis is not supported: If no spatial correlation is observed between macrophage polarization states and drug uptake, this would suggest that macrophage polarization alone does not determine drug distribution in the CT26 tumor model and that other microenvironment factors, such as necrosis, vasculature, or stroma, may have a greater effect. Despite not supporting the hypothesis, this outcome would provide valuable insight into the complexity of CRC tumors and provide preliminary data for future studies examining additional tumor microenvironment components.

Specific Aim 2: To determine whether PD-L1 expression, a functional marker of macrophage-mediated immunosuppression, spatially corresponds to regions of low anti-PD-1 drug uptake in a CT26 CRC xenograft model.

Methods and Rationale:

Using the autoradiography data from Aim 1, the regions of highest and lowest drug uptake will be mapped across additional sections from the same CT26 xenograft specimens, ensuring spatial consistency across both aims. IHC staining for the 10F.9G2 clone will be performed on the same sections to map the spatial expression of PD-L1 across tumor regions. PD-L1 was selected as a marker of interest because it is directly upregulated by M2 macrophages and is a molecular target of anti-PD-1, representing localized regions of resistance [10]. The same sequential autoradiography and methanol fixation protocol from Aim 1 will be applied prior to IHC staining for PD-L1. Using ImageJ, PD-L1 staining intensity will be quantified across the tumor regions and co-registered with the autoradiography drug uptake maps of the corresponding sections to determine whether areas of high PD-L1 expression correlate with regions of low anti-PD-1 uptake and resistance.

This aim builds on Aim 1 by extending beyond macrophage distribution and examining the downstream consequences of M2 macrophage activity. This will provide a greater understanding of how TAM polarization influences therapeutic resistance spatially. The use of the same tissue section and co-registration method ensures consistency between the two aims and minimizes experimental variability [7].

Anticipated results:

If the hypothesis is supported: Regions of high PD-L1 expression would spatially correspond to areas of low drug uptake in CT26 tumors, suggesting that M2-mediated PD-L1 upregulation creates immunosuppressed regions within the tumor that impair therapeutic response. This would provide evidence supporting the spatial relationship between PD-L1 expression and resistance patterns in CRC tumors, and that the immune landscape of the tumor microenvironment is a critical determinant of targeted therapy efficacy in CRC.

If the hypothesis is not supported: If PD-L1 expression does not correlate spatially with regions of low drug uptake in CT26 tumors, it would suggest that PD-L1 expression alone is not sufficient to explain heterogeneous anti-PD-1 drug distribution within the tumor. This outcome would indicate that other resistance mechanisms, such as vascular distribution or necrosis, may contribute to localized resistance in CRC tumors. This would still advance the understanding of the complexity of targeted therapy resistance in tumors and identify other microenvironment components to investigate in future studies.

Potential Problems:

TAMs have high plasticity, so PD-L1 expression may not be exclusively upregulated by M2 macrophages but also by other cell types within the CT26 tumor [11]. To address this, the PD-L1 findings from Aim 2 can be co-registered with the macrophage distribution maps from Aim 1 for a more comprehensive assessment of whether PD-L1 expression spatially coincides with M2-enriched regions or originates from other cellular sources.

Vascular perfusion differences and necrotic regions represent potential confounding factors, since areas of low drug uptake can reflect poor vascular access or cell scarcity rather than macrophage-mediated resistance. H&E staining performed in conjunction with IHC will enable identification and exclusion of necrotic and fibrotic regions from analysis, and vascular density will be considered when interpreting co-registration findings. Future studies incorporating vascular perfusion markers such as CD31 and multiple collection time points would further address these confounders and capture the dynamic evolution of the tumor immune microenvironment over treatment [5].

CT26 is considered an immunorefractory model with limited T-cell infiltration, which may affect the functional interpretation of drug uptake findings and limit generalizability to immunologically hot tumors [10]. While this study focuses on spatial drug distribution rather than T-cell mediated killing, future studies incorporating T-cell activity markers such as IFN-γ production would help establish whether drug penetration into M1-enriched regions translates to functional antitumor responses.

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